13 Integration in Other Coordinate Systems

Choosing coordinates adapted to the region can simplify both the bounds and the integrand. A change of coordinates also changes the area or volume element: substituting for x, y, and z alone is not enough.

13.1 Integration in Polar Coordinates

Definition 13.1: Polar Coordinates and the Area Element

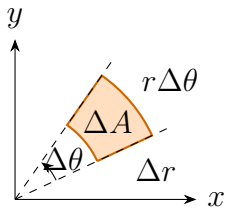
In the plane, write

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r \geq 0.$$

Then $x^2 + y^2 = r^2$, and the area element is

$$\boxed{\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta}.$$

A small polar sector has radial width Δr and arc length approximately $r\Delta\theta$, so $\Delta A \approx r\Delta r\Delta\theta$. Passing to the Riemann-sum limit gives the factor r.



Theorem 13.1: Double Integrals in Polar Coordinates

Suppose f is continuous on a region $\mathcal{D}$ described by

$$\alpha \leq \theta \leq \beta, \qquad r_1(\theta) \leq r \leq r_2(\theta),$$

with nonnegative radial bounds and no overlapping coverage in the interior. Then

$$\iint_{\mathcal{D}} f(x,y)\,\mathrm{d}A = \int_{\alpha}^{\beta}\int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\, r\,\mathrm{d}r\,\mathrm{d}\theta.$$

Example 13.1 (A Right Half-Annulus)**.**

Let $\mathcal{R}$ be the region between $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$ with $x > 0$. Set up an integral for $\iint_{\mathcal{R}} x \, \mathrm{d}A$.

Solution 13.1

The circles become $r = 1$ and $r = 2$. Since $x = r\cos\theta > 0$, we may use

$$1 \le r \le 2, \qquad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}.$$

Including the boundary $x = 0$ does not affect the integral. Thus

$$\begin{aligned}\iint_{\mathcal{R}} x \, \mathrm{d}A &= \int_{-\pi/2}^{\pi/2} \int_1^2 (r\cos\theta) r \, \mathrm{d}r \, \mathrm{d}\theta \\ &= \left(\int_1^2 r^2 \, \mathrm{d}r \right) \left(\int_{-\pi/2}^{\pi/2} \cos\theta \, \mathrm{d}\theta \right) = \frac{7}{3} \cdot 2 = \frac{14}{3}.\end{aligned}$$

Example 13.2 (Population in a Half-Annulus)**.**

A city occupies the same region $\mathcal{R}$, with distances in kilometers. Its population density is $|xy|$ thousand people per square kilometer. Write an iterated integral for the population.

Solution 13.2

The density becomes

$$|xy| = |r^2 \cos\theta \sin\theta|.$$

Consequently, the population in thousands of people is

$$\begin{aligned}P &= \int_{-\pi/2}^{\pi/2} \int_1^2 r^3 |\cos\theta \sin\theta| \, \mathrm{d}r \, \mathrm{d}\theta \\ &= 2 \int_0^{\pi/2} \int_1^2 r^3 \cos\theta \sin\theta \, \mathrm{d}r \, \mathrm{d}\theta \\ &= 2 \cdot \frac{15}{4} \cdot \frac{1}{2} = \frac{15}{4}.\end{aligned}$$

The population is therefore 3750 people. The extra factor r comes from $\mathrm{d}A$, in addition to the r^2 in the density.

13.2 Integration in Cylindrical Coordinates

Definition 13.2: Cylindrical Coordinates and the Volume Element

Cylindrical coordinates combine polar coordinates in the xy-plane with height z:

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z.$$

Here $r \geq 0$ is the distance from the z-axis, and

$$\boxed{\mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z}.$$

A small cylindrical wedge has dimensions approximately Δr, $r\Delta\theta$, and Δz. Hence

$$\Delta V \approx (\Delta r)(r\Delta\theta)(\Delta z).$$

Equivalently, $\mathrm{d}V = \mathrm{d}A\,\mathrm{d}z$ with the polar area element $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$.

Theorem 13.2: Triple Integrals in Cylindrical Coordinates

If a solid $\mathcal{W}$ is described by

$$\alpha \leq \theta \leq \beta, \qquad r_1(\theta) \leq r \leq r_2(\theta), \qquad z_1(r,\theta) \leq z \leq z_2(r,\theta),$$

then, for continuous f,

$$\iiint_{\mathcal{W}} f(x,y,z)\,\mathrm{d}V = \int_{\alpha}^{\beta}\int_{r_1(\theta)}^{r_2(\theta)}\int_{z_1(r,\theta)}^{z_2(r,\theta)} f(r\cos\theta, r\sin\theta, z)\,r\,\mathrm{d}z\,\mathrm{d}r\,\mathrm{d}\theta.$$

The order may be changed when the bounds are rewritten accordingly.

Cylindrical coordinates are useful when boundaries involve x^2+y^2: for instance, $x^2+y^2=a^2$ becomes $r=a$, and $z=x^2+y^2$ becomes $z=r^2$.

Choosing the axis. For a cylinder about the x-axis, use polar coordinates in the yz-plane:

$$y = r\cos\theta, \qquad z = r\sin\theta, \qquad \mathrm{d}V = r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}x.$$

Example 13.3 (Particles Inside a Cylindrical Tube)**.**

A tube is bounded by $y^2 + z^2 = 4$, $x = 0$, and $x = 2$, with distances in centimeters. The stated particle density is $f(x, y, z) = 1 + x^2 + yz$ particles per cubic centimeter. Evaluate the integral of f over the tube.

Solution 13.3

The tube's axis is the x-axis. Use

$$y = r\cos\theta, \qquad z = r\sin\theta,$$

so that the solid is described by

$$0 \le x \le 2, \qquad 0 \le r \le 2, \qquad 0 \le \theta \le 2\pi.$$

Then

$$\begin{aligned} N &= \int_0^2 \int_0^{2\pi} \int_0^2 \big(1 + x^2 + r^2\cos\theta\sin\theta\big) r \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}x \\ &= \int_0^2 \int_0^{2\pi} \big(2(1 + x^2) + 4\cos\theta\sin\theta\big) \, \mathrm{d}\theta \, \mathrm{d}x \\ &= 4\pi \int_0^2 (1 + x^2) \, \mathrm{d}x = 4\pi \left[x + \frac{x^3}{3}\right]_0^2 = \boxed{\frac{56\pi}{3}}. \end{aligned}$$

The yz term integrates to zero because

$$\int_0^{2\pi} \cos\theta\sin\theta \, \mathrm{d}\theta = 0.$$

This also follows from symmetry: reflecting y to $-y$ preserves the tube and reverses the sign of yz.

Modeling note. The lecture's stated density can be negative inside the tube; for instance, $f(0, 1, -\sqrt{2}) = 1 - \sqrt{2} < 0$. Thus $56\pi/3$ is the value of the stated integral, but a physical particle-density model would require a nonnegative function.

13.3 Integration in Spherical Coordinates

Definition 13.3: Spherical Coordinates and the Volume Element

Using ρ for distance from the origin, θ for the angle in the xy-plane, and ϕ for the angle measured down from the positive z-axis,

$$x = \rho\sin\phi\cos\theta, \qquad y = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

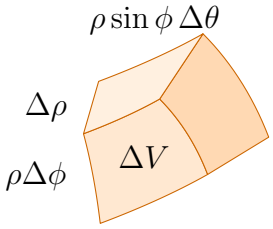
Here $\rho \geq 0$, $0 \leq \theta < 2\pi$, $0 \leq \phi \leq \pi$, and

$$x^2 + y^2 + z^2 = \rho^2, \qquad r = \rho\sin\phi,$$

The volume element is

$$\boxed{\mathrm{d}V = \rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta}.$$

The three local side lengths multiply to give the volume element. The circle traced by changing θ has radius $\rho\sin\phi$, which explains the factor $\sin\phi$.



Theorem 13.3: Triple Integrals in Spherical Coordinates

Let $\mathcal{W}$ be described by

$$\alpha \leq \theta \leq \beta, \qquad \phi_1(\theta) \leq \phi \leq \phi_2(\theta), \qquad \rho_1(\phi,\theta) \leq \rho \leq \rho_2(\phi,\theta).$$

For continuous f,

$$\iiint_{\mathcal{W}} f(x,y,z)\,\mathrm{d}V = \int_\alpha^\beta \int_{\phi_1(\theta)}^{\phi_2(\theta)} \int_{\rho_1(\phi,\theta)}^{\rho_2(\phi,\theta)} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\,\rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta.$$

Example 13.4 (Volume of the Unit Ball)**.**

Let $\mathcal{B}$ be the ball of radius 1 centered at the origin. Find its volume.

Solution 13.4

The ball is $0 \le \rho \le 1$, $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$. Therefore,

$$\begin{aligned}\operatorname{Vol}(\mathcal{B}) &= \int_0^{2\pi}\int_0^{\pi}\int_0^{1} \rho^2 \sin\phi \,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta \\ &= 2\pi\,[-\cos\phi]_0^{\pi}\left[\frac{\rho^3}{3}\right]_0^{1} = \boxed{\frac{4\pi}{3}}.\end{aligned}$$

Example 13.5 (Average Height of a Truncated Cone)**.**

Let $\mathcal{U}$ be the solid inside $z^2 = x^2+y^2$ between $z = 1$ and $z = 2$. Find the average z-coordinate of points in $\mathcal{U}$.

Solution 13.5

Since $z > 0$, the solid lies inside the upper cone. In spherical coordinates,

$$0 \le \theta \le 2\pi, \qquad 0 \le \phi \le \frac{\pi}{4}, \qquad \sec\phi \le \rho \le 2\sec\phi.$$

The radial bounds come from $1 \le \rho\cos\phi \le 2$. Thus

$$\begin{aligned}V &= \int_0^{2\pi}\int_0^{\pi/4}\int_{\sec\phi}^{2\sec\phi} \rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta \\ &= \frac{14\pi}{3}\int_0^{\pi/4}\frac{\sin\phi}{\cos^3\phi}\,\mathrm{d}\phi = \frac{14\pi}{3}\left[\frac{1}{2\cos^2\phi}\right]_0^{\pi/4} = \frac{7\pi}{3}, \\ \iiint_{\mathcal{U}} z\,\mathrm{d}V &= \int_0^{2\pi}\int_0^{\pi/4}\int_{\sec\phi}^{2\sec\phi} (\rho\cos\phi)\rho^2\sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta \\ &= \frac{15\pi}{2}\int_0^{\pi/4}\frac{\sin\phi}{\cos^3\phi}\,\mathrm{d}\phi = \frac{15\pi}{4}.\end{aligned}$$

Hence

$$z_{\text{avg}} = \frac{1}{V}\iiint_{\mathcal{U}} z\,\mathrm{d}V = \frac{15\pi/4}{7\pi/3} = \boxed{\frac{45}{28}}.$$

13.4 Choosing a Coordinate System

The same solid can admit several useful descriptions. For the truncated cone above, cylindrical coordinates give the simpler bounds

$$1 \le z \le 2, \qquad 0 \le r \le z, \qquad 0 \le \theta \le 2\pi.$$

As a check,

$$V = \int_1^2 \int_0^{2\pi} \int_0^z r \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}z = \pi \int_1^2 z^2 \, \mathrm{d}z = \frac{7\pi}{3},$$

$$\iiint_{\mathcal{U}} z \, \mathrm{d}V = \int_1^2 \int_0^{2\pi} \int_0^z zr \, \mathrm{d}r \, \mathrm{d}\theta \, \mathrm{d}z = \pi \int_1^2 z^3 \, \mathrm{d}z = \frac{15\pi}{4}.$$

Both methods yield $z_{\text{avg}} = 45/28$.

Coordinates	**Typical geometry**	**Area/volume factor**
Polar	Disks, annuli, sectors	r
Cylindrical	Cylinders, axial symmetry	r
Spherical	Balls, spherical shells, cones	$\rho^2 \sin\phi$

Setting up an integral.

1. Sketch the region and identify its axis or symmetry.
2. Rewrite all boundary equations in the chosen coordinates.
3. Choose an integration order and determine bounds from outside inward.
4. Replace every occurrence of the original coordinates in the integrand.
5. Include the area or volume factor exactly once, then evaluate from inside outward.

Common pitfalls. In spherical coordinates, ρ is distance from the origin, whereas $r = \rho \sin\phi$ is distance from the z-axis. The angle ϕ is measured from the positive z-axis, so the upper half-space has $0 \le \phi \le \pi/2$. Bounds must cover the region once; splitting a region into pieces may be necessary.